BCSE302P — DATABASE SYSTEMS LAB

COMPLETE PROJECT REPORT

POLICY INTELLIGENCE FOR AGRICULTURE

Analysis of Farmer Feedback on Government Welfare Schemes using Natural Language Processing and Large Language Models



Rice farms, Chittoor district, Andhra Pradesh. Source: Wikimedia Commons · rajaraman sundaram · CC BY 3.0

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1. EXECUTIVE SUMMARY

Government welfare schemes for farmers, such as PM-KISAN, Rythu Bharosa, crop insurance and the Kisan Credit Card, are commonly evaluated through satisfaction ratings. Farmer surveys, however, also record open-ended responses describing what was beneficial and what failed. These written responses contain detailed evidence but are rarely analysed systematically.

This project analyses a field survey of **1,500 farmers from Chittoor district, Andhra Pradesh**, verified by the Ethnic Community, Government of Andhra Pradesh. The survey was converted into a normalised, rule-validated database, and the 1,214 written responses were analysed using natural language processing (NLP) and large language models (LLMs). Eleven sentiment models were evaluated against a human-labelled test set to determine which approach interprets farmer feedback most accurately.

1,500	**22**	**40,286**	**11**	**0.765**
farmers	database tables	records	models evaluated	best macro-F1

Key Findings

- **Awareness does not translate into enrolment:** 89.4% of farmers aware of Rythu Bharosa are enrolled, compared with 9.1% for Crop Insurance.
- **Ratings conceal service problems:** for timeliness and enrolment, ratings contradict the farmers' own written comments in approximately 29% of cases.
- **Enrolment is the weakest aspect** of all four major schemes.
- **General-purpose sentiment tools perform poorly:** the VADER lexicon achieved a macro-F1 of 0.223, whereas Fine-tuned ModernBERT-base, trained on labelled farmer responses, achieved 0.765.
- **Domain knowledge improves LLM performance:** the Indian agriculture LLM Dhenu2 achieved 0.704, compared with 0.391 for a general LLM of identical size.

The findings of this project are being prepared for publication as a research paper targeting a Q1 journal, under the guidance of Prof. Sangeetha S.

2. PROBLEM STATEMENT

2.1 Background

Agriculture in Chittoor district is predominantly small-scale. More than half of the surveyed farmers hold less than 2.5 acres, and the principal crops are mango, paddy and sugarcane. For such households, scheme benefits directly influence whether inputs such as seed and fertiliser can be purchased on time.



Figure 2.1: Traditional ploughing in Andhra Pradesh

Source: Wikimedia Commons · PJeganathan · CC BY-SA 3.0



Figure 2.2: Farmland at Gudimallam, Chittoor district

Source: Wikimedia Commons · Ms Sarah Welch · CC0

In Andhra Pradesh, many schemes are delivered through village-level service centres known as **Rythu Bharosa Kendrams**, where farmers register, seek information and receive support. The effectiveness of a scheme is conventionally measured through farmer satisfaction ratings.



Figure 2.3: A Rythu Bharosa Kendram, the village-level service centre for farm schemes

Source: Wikimedia Commons · Veera Ram Kumar · CC0

2.2 Problem Definition

The survey data was available only as a single spreadsheet, which presented four significant problems:

Problem	Description	Consequence
Inconsistent naming	a single scheme appeared under 93 different spellings	scheme-level counts could not be computed reliably
Multi-valued cells	several crops or schemes recorded within one cell	the data could not be searched or aggregated
Absence of validation	no mechanism prevented impossible values, such as farming experience exceeding age	erroneous values could enter the analysis
Unanalysed text	1,214 farmers provided written feedback	the most detailed evidence remained unused

Table 1: Problems identified in the original survey data

Research question: How can the survey be stored in a clean, reliable and queryable form, and how can farmers' written feedback be analysed with sufficient accuracy to inform policy?

2.3 Motivation

When satisfaction ratings indicate that a scheme is performing well while written feedback describes delays and documentation barriers, policy decisions risk being based on an incomplete signal. A well-designed database ensures that every reported figure is traceable to an individual interview, and validated NLP ensures that automated text analysis is reliable before it is used to support conclusions.

3. OBJECTIVES, SCOPE AND REQUIREMENT ANALYSIS

3.1 Objectives

No.	Objective	Status
1	Design a normalised database for the survey with integrity rules that prevent invalid data	Completed
2	Load all 1,500 interviews while preserving traceability to the original records	Completed
3	Analyse written feedback using NLP and store each result together with its source model	Completed
4	Answer policy-relevant questions directly from the stored data	Completed
5	Identify the most accurate model for farmer feedback through evaluation against human labels	Completed
6	Generate reports, a policy brief and a research paper draft	Completed

Table 2: Project objectives

3.2 Scope

Within scope	Outside scope
All 1,500 interviews from Chittoor district	Applications for collecting new survey data
Database design, implementation, loading and testing	Other districts and states
NLP: sentiment, aspect-based sentiment, topics, LLMs and model comparison	Real-time or streaming data
Reports, policy brief and dashboard dataset	Cloud deployment
A human-labelled evaluation set	Publication of farmers' personal information

Table 3: Project scope

3.3 Requirement Analysis

Requirement	Description
Data to be stored	farmer profiles, villages, crops, schemes and their spellings, awareness and enrolment, ratings of six service aspects, written feedback, model outputs, quality flags and human labels
System users	the researcher, annotators labelling responses, policy makers reviewing reports, and faculty verifying results
Operations required	bulk loading from Excel, cleaning and validation, storage of model outputs, text search, and complete reconstruction from the source file
Information to be analysed	awareness versus enrolment by scheme, sentiment by scheme and aspect, disagreement between ratings and comments, the weakest aspect of each scheme, and model accuracy

Table 4: Requirement analysis

Functional Requirements

ID	Requirement
FR1	Import the raw survey without modification
FR2	Clean the data and flag contradictory responses without deleting them
FR3	Store the data in related tables with rules that reject invalid values
FR4	Store every model output together with the model that produced it
FR5	Record human labels for model evaluation
FR6	Answer policy questions
FR7	Trace any record to its original row
FR8	Generate reports, dashboard data and a paper draft

Table 5: Functional requirements

Data Requirements

Survey section	Questions	Content
A — Profile	12	age, gender, village, education, land, crops, income, experience
B — Awareness	5	schemes known, schemes joined, information sources
C — Feedback	11	satisfaction, six aspect ratings, three written responses
D — Outlook	4	government support, trend, confidence, principal challenge
E — Interview	4	language, comfort, observations, duration

Table 6: Data requirements by survey section

4. TECHNOLOGY STACK

All tools used are freely available, and the complete system runs on a single laptop.

Layer	Technology	Role in the project
Database	PostgreSQL 16 (relational), pgAdmin	storage of survey data, model outputs and labels in 22 related tables
Programming language	Python 3.12	complete pipeline of 12 scripts and 21 modules
Data handling	pandas, openpyxl, RapidFuzz	reading Excel files, cleaning, fuzzy matching of scheme names
Database connectivity	psycopg 3	parameterised loading within a single transaction
Classical NLP	NLTK VADER, scikit-learn (LDA, TF-IDF)	lexicon sentiment, topic modelling, trained baseline
Transformer NLP	Hugging Face Transformers, SetFit	RoBERTa, DeBERTa-v3, ModernBERT and few-label learning
Large language models	Dhenu2 (KissanAI), Qwen2.5-3B, Claude	classification of feedback using the labelling guidelines
GPU runtime	PyTorch (CUDA), bitsandbytes 4-bit, NVIDIA RTX 2050	execution of 3-billion-parameter models on a 4 GB GPU
Statistics	SciPy	chi-square and McNemar tests, bootstrap confidence intervals
Reporting	python-docx, Power BI dataset	statistical report, paper draft and dashboard data
Development tools	VS Code, Git and GitHub, Google Colab	development, version control, optional cloud GPU

Table 7: Technology stack

4.1 Database Selection and Justification

The survey has a fixed structure, and its most important information lies in the relationships between entities: a farmer may know many schemes, a scheme involves many farmers, and each feedback record contains six aspect ratings. A relational database represents these relationships directly and enforces data integrity at the point of entry. PostgreSQL was therefore selected.

Option	Type	Decision
PostgreSQL	Relational	Selected — relationships, integrity constraints, transactions, full-text search, open source
MongoDB	Document	Not selected — the survey has a fixed structure that requires relationships between records
Neo4j	Graph	Not selected — relationships are only one or two levels deep
Redis	Key-value	Not selected — not designed for analytical queries
BigQuery	Cloud warehouse	Not selected — unnecessary cost and scale for 1,500 records

Table 8: Database technologies considered

5. PROJECT FOLDER STRUCTURE

The project is organised into dedicated folders, and the original survey file is never modified.



Figure 5.1: Project folder structure

Folder	Contents
data/	original survey file (unmodified)
db/	database schema and analysis queries
src/	21 reusable Python modules
scripts/	12 numbered pipeline steps
notebooks/	Google Colab notebook for GPU execution
output/	generated outputs, including labelling sheets
results/	aggregate model comparison results
docs/	project report, presentation and figures
reports/	statistical report, paper draft and policy brief

Table 9: Purpose of each folder

Folders containing personal information (data/ and output/) are excluded from the public repository.

6. SYSTEM WORKFLOW

The workflow converts a single Excel file into validated analytical outputs through five stages.

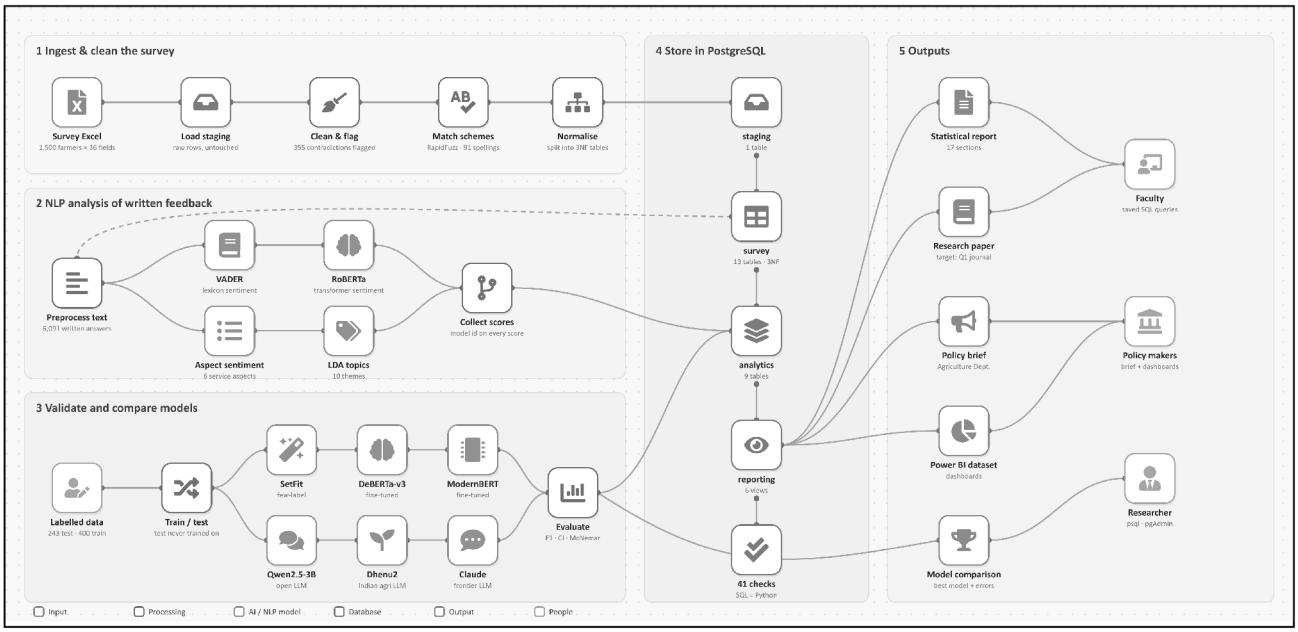


Figure 6.1: End-to-end project workflow

Stage	Process	Output
1. Ingestion and cleaning	the source file is copied unchanged and then cleaned: scheme names are matched, lists are separated and contradictions are flagged	1,500 clean, traceable records
2. NLP analysis	four models analyse the written responses for overall sentiment, aspect-level sentiment and recurring topics	11,151 model outputs
3. Validation and comparison	labelled responses are divided into test and training sets; eleven models are trained or prompted and evaluated	a statistically tested model ranking
4. Storage	all data is loaded into four database layers in one transaction, followed by 41 automated checks	a single verified data source
5. Outputs	reports, policy brief, dashboard data and the paper draft are generated from the stored data	outputs for each user group

Table 10: Workflow stages

6.1 Data Management Principles

Two principles govern data handling: the source file is never edited, and human labels are never overwritten. Consequently, the complete system can be rebuilt from the source file in approximately 14 seconds with identical results.

7. SYSTEM ARCHITECTURE

The architecture follows the sequence: users, application, database and outputs.

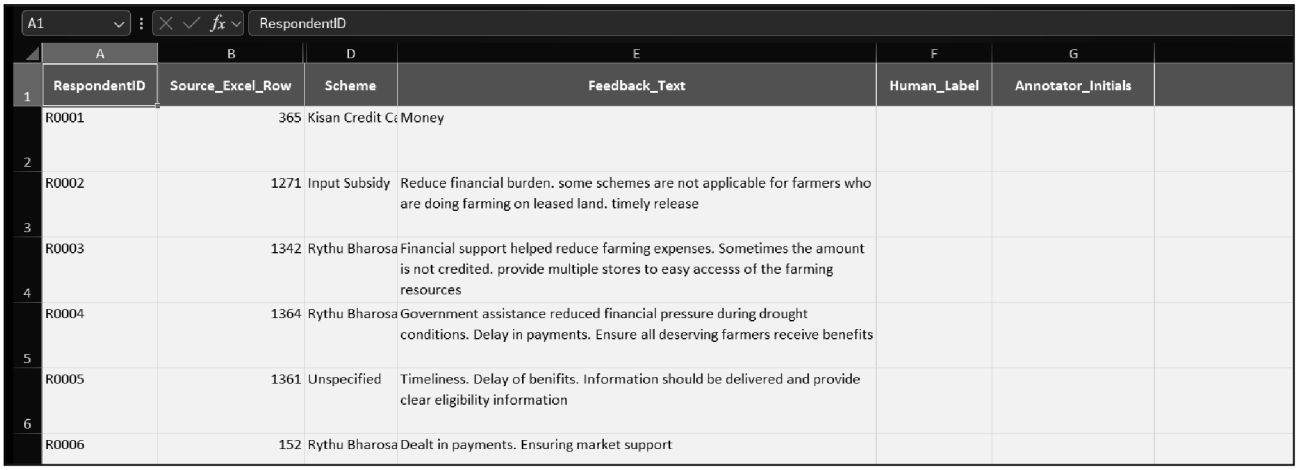


Figure 7.1: System architecture

The inputs are the survey file and the human labels. The Python application cleans the data, executes the NLP models and statistical tests, and loads the database within a single transaction, which prevents partially loaded states. The database is organised into four layers. After each load, the application reads the results back and verifies that the database and the analysis are consistent. All outputs are generated from the database.

User	Interaction with the system
Researcher	executes the pipeline and inspects data through pgAdmin
Annotators	label responses in Excel worksheets using single-key entries (1 = Positive, 2 = Neutral, 3 = Negative)
Policy makers	review the policy brief and dashboard dataset
Faculty	verify any result against the original survey record

Table 11: System users



A1 | RespondentID

	A	B	D	E	F	G
1	RespondentID	Source_Excel_Row	Scheme	Feedback_Text	Human_Label	Annotator_Initials
2	R0001	365	Kisan Credit Ca	Money		
3	R0002	1271	Input Subsidy	Reduce financial burden. some schemes are not applicable for farmers who are doing farming on leased land. timely release		
4	R0003	1342	Rythu Bharosa	Financial support helped reduce farming expenses. Sometimes the amount is not credited. provide multiple stores to easy accesss of the farming resources		
5	R0004	1364	Rythu Bharosa	Government assistance reduced financial pressure during drought conditions. Delay in payments. Ensure all deserving farmers receive benefits		
6	R0005	1361	Unspecified	Timeliness. Delay of benifits. Information should be delivered and provide clear eligibility information		
	R0006	152	Rythu Bharosa	Dealt in payments. Ensuring market support		

Figure 7.2: Labelling worksheet used by annotators (participant names hidden)

8. DATABASE DESIGN

8.1 Database Layers

Layer	Objects	Purpose
Staging	1 table	raw survey as received
Survey	13 tables	cleaned and normalised survey data
Analytics	9 tables	model outputs, topics, quality flags and human labels
Reporting	6 views	pre-defined answers to policy questions

Table 12: Database layers

8.2 Entity-Relationship Diagrams

The following diagrams were generated directly from the implemented database. Primary keys (PK) identify each row uniquely, foreign keys (FK) reference related tables, and the crow's-foot notation denotes the "many" side of a relationship.

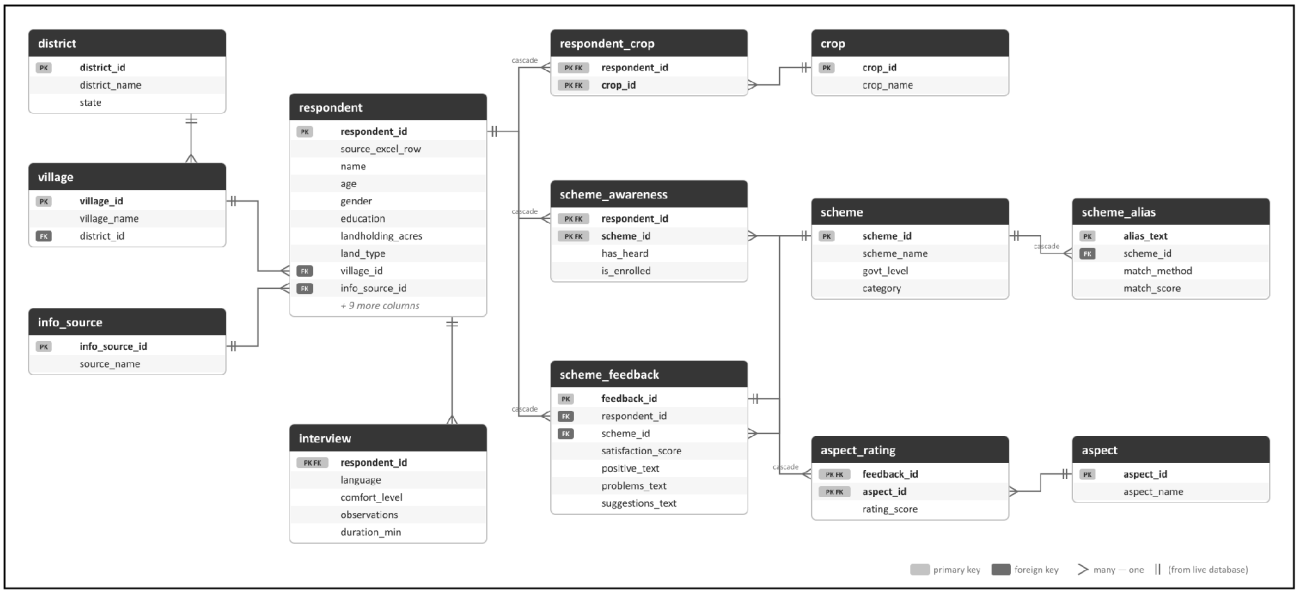


Figure 8.1: ER diagram of the survey layer

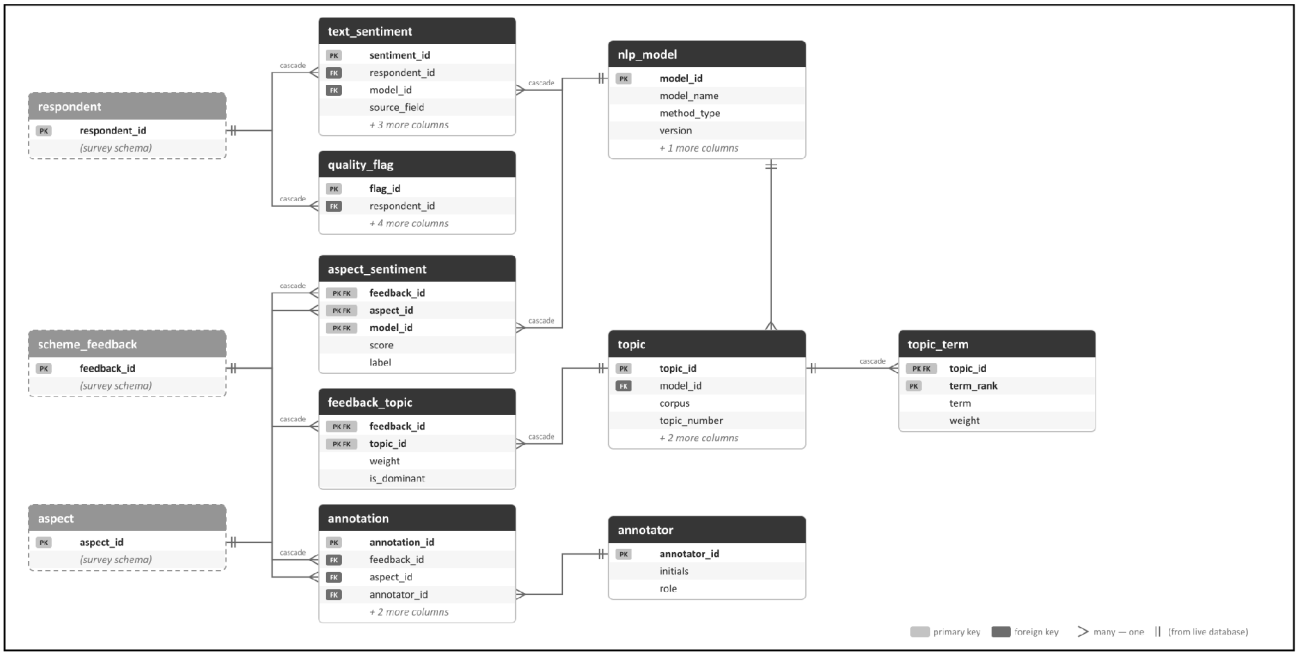


Figure 8.2: ER diagram of the analytics layer

8.3 Normalisation

In the original spreadsheet, each row contained multi-valued cells, six repeating rating columns and district information duplicated for every farmer in a village. Multi-valued attributes were moved to associative tables to satisfy First Normal Form, and attributes dependent on entities other than the farmer, such as the district of a village, were moved to separate tables to satisfy Second and Third Normal Form. As a result, every fact is stored exactly once.

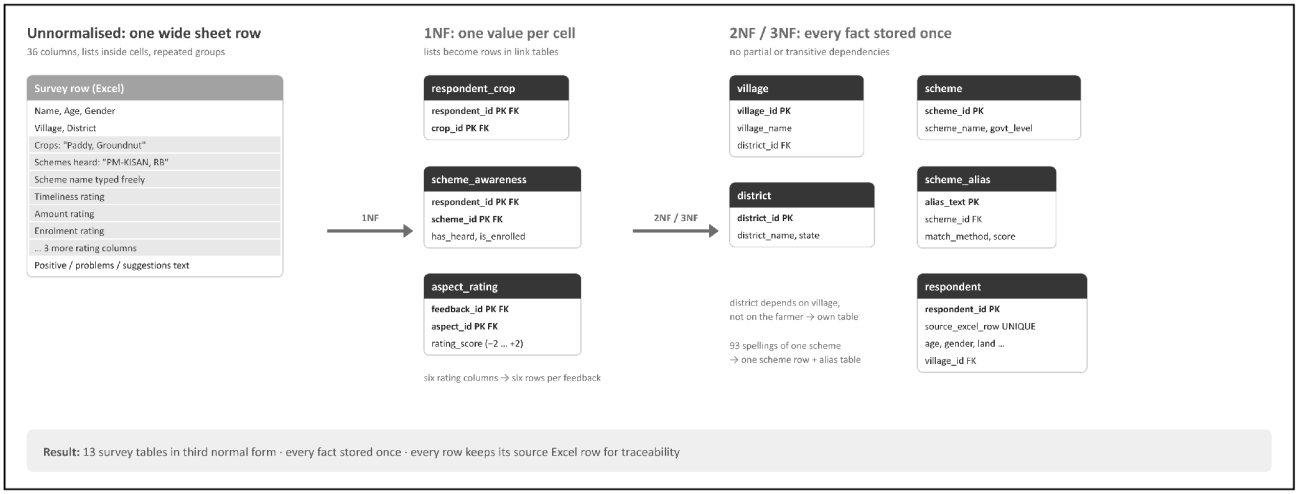


Figure 8.3: Normalisation from the spreadsheet to Third Normal Form

8.4 Keys and Integrity Constraints

Constraint type	Count	Guarantee
Primary key	23	each row is uniquely identified
Foreign key	26	every reference points to an existing record
Unique	13	duplicates are prevented, e.g. one feedback per farmer per scheme
Check	25	values remain within valid ranges, e.g. ratings between −2 and +2
Index	43	efficient retrieval, including full-text search

Table 13: Keys and integrity constraints

Deletion behaviour was defined explicitly. Shared reference data, such as schemes and crops, cannot be deleted while referenced. Records belonging to a farmer are deleted together with that farmer (16 cascading relationships), so that a withdrawal of consent removes all associated data in a single operation.

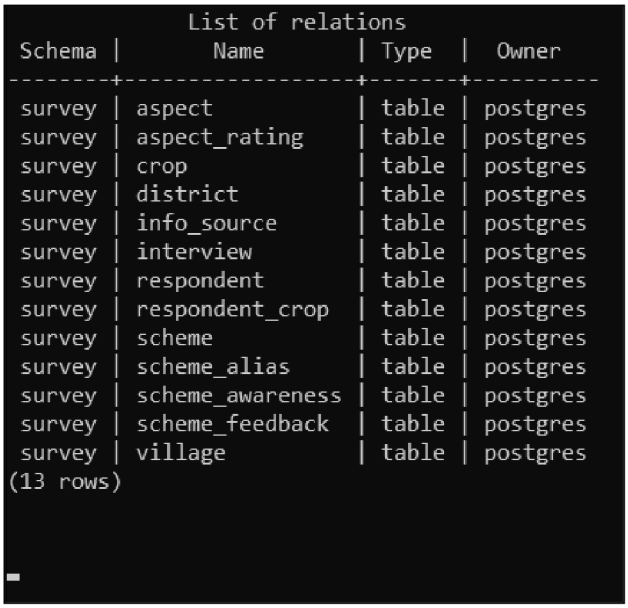


Figure 8.4: Survey tables in the implemented database

9. NLP MODELS, LARGE LANGUAGE MODELS AND MODULES

9.1 System Modules

Step	Module	Function
01	Pipeline	cleaning, scheme matching, VADER and RoBERTa sentiment
02	Analysis	data-quality checks, statistical tests, topic modelling
03	Dashboard data	Power BI dataset
04	Test worksheets	stratified 20% sample for human labelling
05	Evaluation	accuracy of stored models against human labels
06	Statistics	descriptive tables for reporting
07	Database load	database construction and 41 verification checks
08–10	Reports	statistical report, paper draft and policy brief
11	Training worksheets	400 training responses: uncertainty-based and random selection
12	Model comparison	training, prompting and evaluation of all eleven models

Table 14: System modules

9.2 Types of NLP

NLP type	Model(s)	Method
Lexicon-based sentiment	VADER with agricultural lexicon	aggregates the polarity of known words
Pre-trained transformer	RoBERTa (Twitter)	neural sentiment model applied without further training
Aspect-based sentiment	keyword method; DeBERTa ABSA	sentiment towards each service aspect separately
Topic modelling	Latent Dirichlet Allocation	identification of recurring themes in complaints
Few-label learning	SetFit	contrastive learning from a few hundred examples
Fine-tuning	DeBERTa-v3, ModernBERT	full training of a transformer on farmer responses
LLM prompting	Dhenu2, Qwen2.5, Claude	classification guided by the labelling instructions, with zero or three examples

Table 15: Types of NLP used

9.3 Large Language Models

Model	Developer	Parameters	Reason for inclusion
Dhenu2	KissanAI (India)	3 billion	domain-specific model for Indian agriculture
Qwen2.5-3B-Instruct	Alibaba	3 billion	general-purpose model of identical size, for controlled comparison
Claude Opus 5.5	Anthropic	frontier-scale	state-of-the-art reference model
ModernBERT, DeBERTa-v3	Answer.AI, Microsoft	approx. 150 million	encoder models suitable for full fine-tuning

Table 16: Large language models and encoders evaluated

The 3-billion-parameter models were loaded with 4-bit quantisation to fit within the 4 GB memory of the available GPU.

9.4 Evaluation Design

Dataset	Size	Purpose
Test set	243 responses	evaluation only; never used for training
Training set	400 responses	training of trainable models and selection of few-shot examples
Aspect test set	131 responses (274 aspect mentions)	evaluation of aspect-level models

Table 17: Evaluation datasets

Labels were produced jointly by the author and one additional annotator. For the training set, labels were first suggested by an AI model and then verified by the annotators, who changed 86 of 400 of the suggestions. Each trainable model was executed with five random seeds. Results are reported with 95% bootstrap confidence intervals, and differences between models were tested using the exact McNemar test.

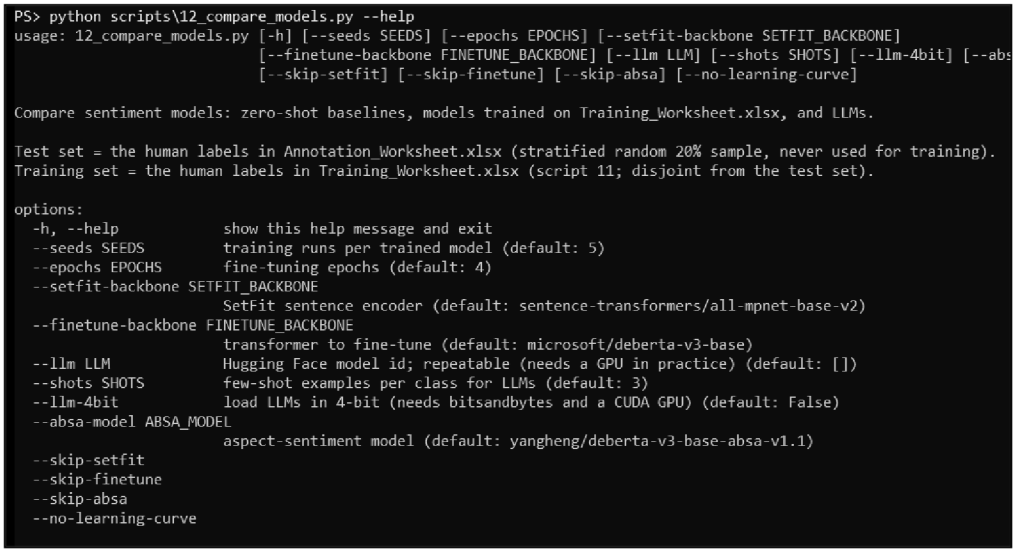


Figure 9.1: Model comparison module and its default models

10. STATISTICAL ANALYSIS OF THE DATASET

51 years	2.0 acres	20 years	73.7%	90.5%
mean age	median landholding	mean farming experience	aware of schemes	interviews in Telugu

10.1 Respondent Profile

Gender	Farmers	Share
Male	989	67.2%
Female	483	32.8%

Landholding	Farmers	Share
Marginal (<2.5 ac)	581	53.3%
Small (2.5-5 ac)	246	22.5%
Semi-medium (5-10 ac)	136	12.5%
Large (10+ ac)	128	11.7%

Table 18: Gender and landholding distribution

Education	Farmers	Share
High school (9-10)	288	19.6%
Primary school (1-5)	275	18.7%
No formal education	263	17.9%
Intermediate (11-12)	229	15.6%
Graduate	222	15.1%
Middle school (6-8)	125	8.5%

Table 19: Education distribution

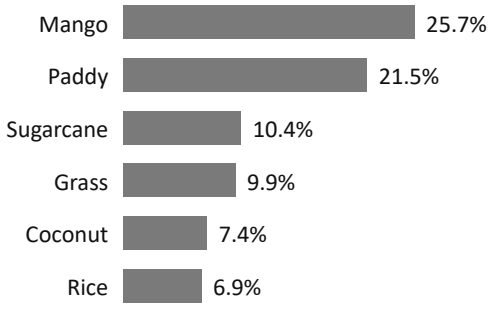


Figure 10.1: Principal crops (% of farmers)

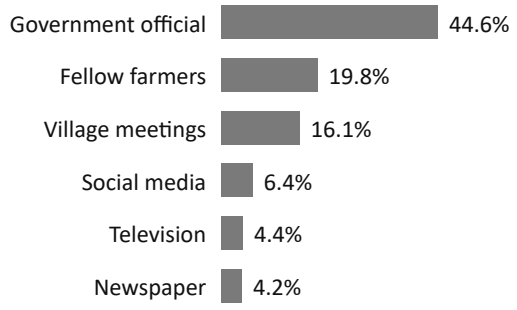


Figure 10.2: Sources of scheme information (%)

10.2 Sentiment and Themes

The initial lexicon model classified 55.6% of written feedback as positive, 21.5% as neutral and 22.9% as negative. As reported in Chapter 11, this model was subsequently found to be unreliable, which motivated the evaluation of more accurate models.

Complaint theme	Share	Representative terms
Payment delays & bank verification	22%	payment, delayed, bank, delay, eligible, family
Online registration & documents	18%	application, online, registration, document, required, delay
Insufficient support for crop needs	17%	need, insufficient, crop, official, support, requirement
Late credit of benefits	22%	time, delay, sufficient, receiving, late, credited
Low awareness & slow office approval	21%	time, application, lack, long, difficulty, awareness

Table 20: Principal complaint themes identified by topic modelling

Rank	Suggestion theme
1	Increase the benefit amount
2	Simplify / speed up the application process
3	Run more awareness programs
4	Ensure faster, more reliable disbursement

Table 21: Most frequent improvement suggestions

10.3 Association Between Sentiment and Respondent Characteristics

Chi-square tests indicate that all characteristics below are significantly associated with sentiment ($p < 0.05$). Cramér's V values between approximately 0.10 and 0.15 indicate that these associations are weak.

Characteristic	Chi-square	p-value	Cramér's V
Education	58.53	9.2e-07	0.143
Landholding	32.89	1.1e-05	0.125
Info Source	27.42	3.7e-02	0.107
Scheme	42.32	1.0e-03	0.133
Gender	20.37	3.8e-05	0.119
Income	48.2	5.7e-07	0.13

Table 22: Chi-square tests of sentiment against respondent characteristics

11. NLP RESULTS

All eleven models were evaluated on the same 243 test responses, none of which were used during training. Macro-F1 is used as the primary metric because it weights positive, neutral and negative classes equally and is therefore not inflated by the dominant class.

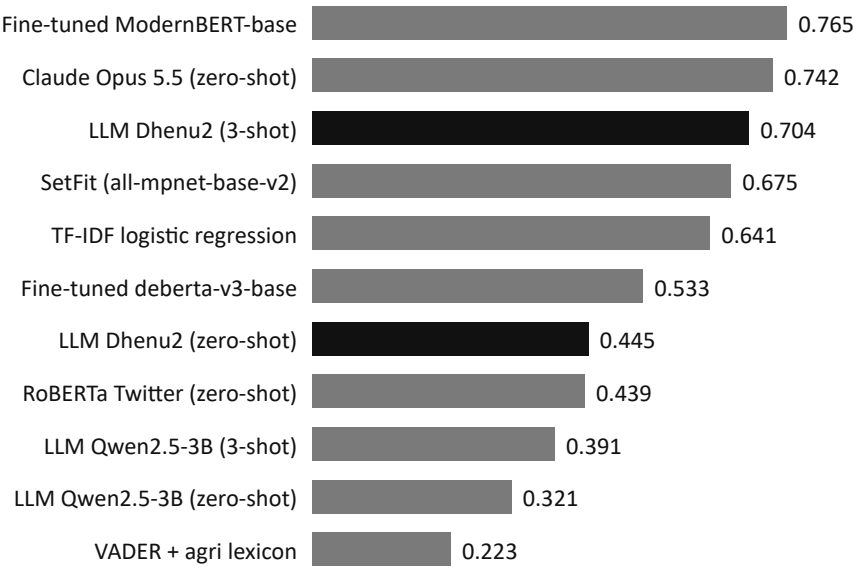


Figure 11.1: Macro-F1 of all evaluated models (Dhenu2 shown in black)

Model	Type	Accuracy	Macro-F1	95% CI	Mean ± SD (5 seeds)
Fine-tuned ModernBERT-base	trained (fine-tuned)	0.835	0.765	0.687 - 0.832	0.750 ± 0.034
Claude Opus 5.5 (zero-shot)	LLM (prompted, external)	0.786	0.742	0.676 - 0.801	—
LLM Dhenu2-In-Llama3.2-3B-Instruct (3-shot)	LLM (prompted)	0.749	0.704	0.63 - 0.766	—
SetFit (all-mpnet-base-v2)	trained (few-label)	0.840	0.675	0.592 - 0.752	0.685 ± 0.022
TF-IDF logistic regression	trained baseline	0.745	0.641	0.562 - 0.718	—
Fine-tuned deberta-v3-base	trained (fine-tuned)	0.749	0.533	0.476 - 0.595	0.584 ± 0.080
LLM Dhenu2-In-Llama3.2-3B-Instruct (zero-shot)	LLM (prompted)	0.519	0.445	0.38 - 0.51	—
RoBERTa Twitter (zero-shot)	zero-shot baseline	0.506	0.439	0.367 - 0.512	—
LLM Qwen2.5-3B-Instruct (3-shot)	LLM (prompted)	0.457	0.391	0.317 - 0.458	—
LLM Qwen2.5-3B-Instruct (zero-shot)	LLM (prompted)	0.300	0.321	0.268 - 0.376	—
VADER + agri lexicon	zero-shot baseline	0.235	0.223	0.163 - 0.279	—

Table 23: Performance of all models on the test set

Best model: Fine-tuned ModernBERT-base, with a macro-F1 of 0.765 and an accuracy of 83.5%. Its advantage over Claude Opus 5.5 (zero-shot), SetFit (all-mpnet-base-v2) is not statistically significant; these models are therefore regarded as jointly best. It is significantly better than all other models.

11.1 Per-Class Performance of the Best Model

Class	Precision	Recall	F1	Test responses
Positive	0.873	0.724	0.791	76
Neutral	0.833	0.944	0.885	143
Negative	0.722	0.542	0.619	24

Table 24: Per-class performance of the best model

Performance is highest for neutral and positive feedback. Negative feedback is the most difficult class, as only 24 of the test responses were negative.

11.2 Effect of Training Data Size

Training responses	Macro-F1 (TF-IDF model)
100	0.494
200	0.567
300	0.633
400	0.641

Table 25: Learning curve

Performance increased with each additional 100 labelled responses, indicating that further labelling would yield further improvement.

11.3 Aspect-Level Sentiment

Method	Aspect mentions detected	Accuracy	Macro-F1
Keyword ABSA + VADER (current pipeline)	201 of 274	0.343	0.373
DeBERTa ABSA (deberta-v3-base-absa-v1.1)	274 of 274	0.522	0.433

Table 26: Aspect-level sentiment evaluation

The dedicated DeBERTa aspect model processed every aspect mention and outperformed the original keyword-based method.

12. RESEARCH FINDINGS

Finding 1: Awareness does not translate into enrolment

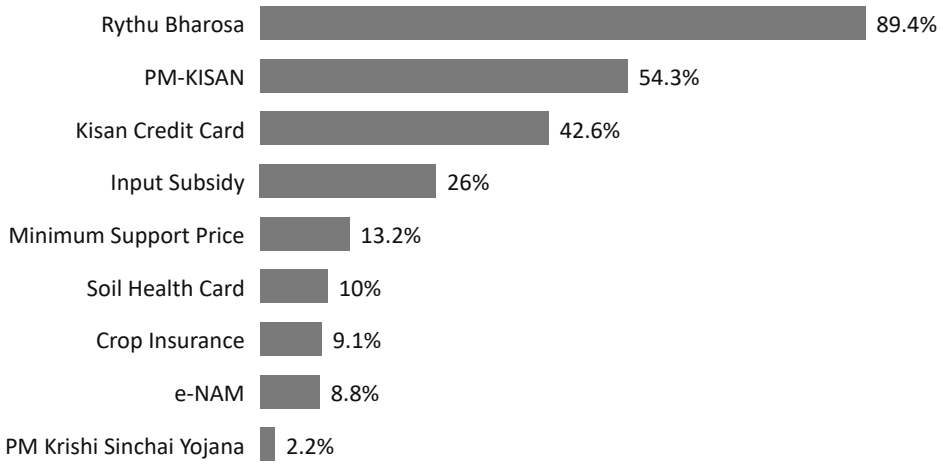


Figure 12.1: Enrolment among farmers aware of each scheme (%)

Of the farmers aware of Rythu Bharosa, 89.4% are enrolled; of the 143 farmers aware of Crop Insurance, only 9.1% are enrolled. For crop insurance, the principal barrier is therefore enrolment rather than awareness.

Finding 2: Ratings and written feedback frequently disagree

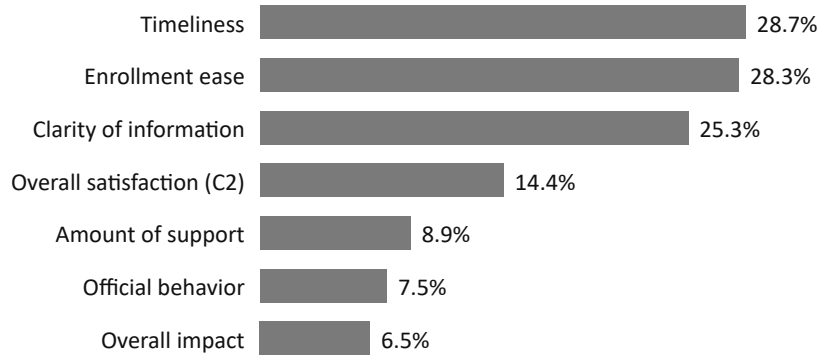


Figure 12.2: Rate of disagreement between ratings and written feedback, by aspect (%)

Disagreement is highest for timeliness (28.7%) and enrolment (28.3%). A McNemar–Bowker test confirms that the overall disagreement is statistically significant ($\chi^2 = 192.73$, $p < 0.001$). Satisfaction ratings alone therefore conceal substantive service problems.

Finding 3: Enrolment is the weakest aspect of every major scheme

Scheme	Weakest aspect	Negative mentions
Input Subsidy	Enrolment	53.8%
Kisan Credit Card	Enrolment	50.0%
PM-KISAN	Enrolment	47.6%
Rythu Bharosa	Enrolment	40.6%

Table 27: Weakest service aspect of each major scheme

The four major schemes share the same weakness, which indicates a common registration process requiring improvement rather than four unrelated problems.

Finding 4: General-purpose sentiment tools are unreliable for farmer feedback

The VADER lexicon, which remains widely used in applied research, achieved a macro-F1 of 0.223, and RoBERTa trained on social-media text achieved 0.439. Models trained on labelled farmer responses reached 0.765. Studies that apply such tools to farmer surveys without validation may therefore reach incorrect conclusions.

Finding 5: Agricultural domain knowledge improves LLM performance

With three examples, the Indian agriculture LLM Dhenu2 achieved a macro-F1 of 0.704, whereas Qwen2.5-3B, a general model of identical size, achieved 0.391. Domain-specific training substantially improves the interpretation of farmer feedback.

> Findings 1 to 3 are derived directly from survey responses; Findings 4 and 5 are derived from the model evaluation. Together they constitute the core contributions of the research paper in preparation.

13. TESTING AND PERFORMANCE

13.1 Automated Verification

After every load, 41 automated checks compare the database contents with the results computed independently in Python, covering row counts, sentiment totals, disagreement rates, scheme-level figures and awareness counts. All 41 checks pass.

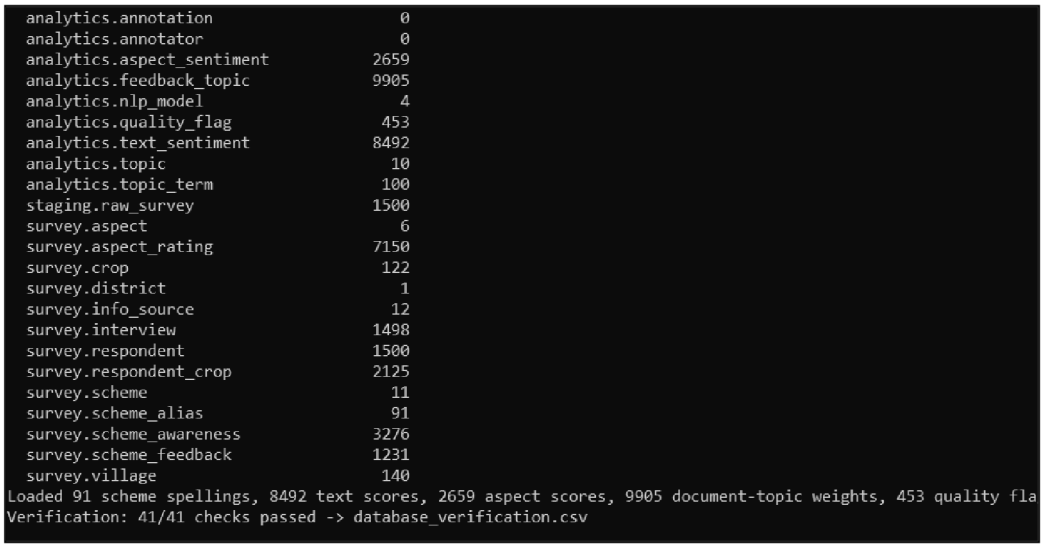


Figure 13.1: Database load with row counts and the result "Verification: 41/41 checks passed"

13.2 Integrity Constraint Testing

Invalid input attempted	Constraint applied	Result
rating of 5 (valid range −2 to +2)	check constraint	rejected
feedback for a non-existent farmer	foreign key	rejected
duplicate farmer identifier	primary key	rejected
sentiment label "Happy"	check constraint	rejected

Table 28: Results of deliberate invalid-data tests

Responses that are internally contradictory but genuine, for example a farmer reporting no awareness of schemes while rating one, are retained and flagged (453 flags), since deleting genuine responses would introduce bias.

13.3 Performance Observations

Measure	Value
Complete rebuild from the source file	13.7 seconds
Execution time of any reporting query	below 15 milliseconds
Database size	16 MB
Fine-tuning time per model on the GPU	approximately 33 seconds

Table 29: Performance observations

14. CHALLENGES AND SOLUTIONS

Category	Challenge	Solution
Data handling	one scheme recorded under 93 different spellings	fuzzy matching, with every spelling and match method recorded in a reference table
Data handling	several responses stored in one cell	separation using the questionnaire option list into associative tables
Data handling	355 farmers with contradictory responses	responses retained and flagged rather than deleted
Data handling	responses such as "no problems" scored as negative	such non-informative responses treated as empty
Database design	six repeating rating columns	restructured into one row per feedback and aspect
Database design	outputs from multiple models	a model reference table; each score stores its model identifier
Database design	consistency between database and analysis	41 automated checks after every load
Technical	LLMs exceeding available memory	4-bit quantisation on the GPU
Technical	GPU memory exhaustion between models	explicit memory release and cached predictions enabling resumption
Methodological	interviews in Telugu recorded in English	English models used; documented as a limitation
Methodological	topic models varying between runs	topic results reported as exploratory

Table 30: Challenges encountered and solutions implemented

15. CONCLUSION AND FUTURE SCOPE

15.1 Achievements

- A normalised database of 22 related tables containing 40,286 records, with 26 foreign keys and 25 check constraints, rebuilt and verified automatically.
- An NLP layer in which every output is stored with its source model, and a statistically tested comparison of 11 models, including an Indian agriculture LLM.
- Evidence-based policy findings: the awareness–enrolment gap, the disagreement between ratings and written feedback, and the common weakness in enrolment.
- Reports, a policy brief, a dashboard dataset and a research paper draft generated from the stored data.

15.2 Learning Outcomes

- Transformation of unstructured real-world data into a normalised relational design.
- Practical application of keys, integrity constraints and transactions.
- Secure connectivity between Python and a relational database.
- Rigorous model evaluation using held-out data, repeated runs, confidence intervals and significance tests.
- Efficient execution of large language models on limited hardware.

15.3 Limitations

- The data covers a single district and a single survey round.
- Interviews were conducted in Telugu and recorded in English, which may reduce nuance.
- Labels were produced jointly as a single consensus set; inter-annotator agreement could therefore not be measured.
- Negative feedback was under-represented in the test set (24 responses), which increases uncertainty for that class.

15.4 Future Scope

Research publication: The findings of this project are currently being written as a research paper **targeting a Q1 journal**, under the guidance of Prof. Sangeetha S. The principal contributions are the measured gap between farmer ratings and written feedback, the awareness–enrolment gap, and a validated comparison of NLP models including an Indian agriculture LLM.

- Expansion of the labelled dataset with independent double annotation.
- Recomputation of aspect-level findings using the best-performing model.
- A live Power BI dashboard connected directly to the database.
- Extension to additional districts and survey rounds, with cloud deployment.
- Evaluation of models operating directly on Telugu text.

15.5 Project Resources

Resource	Link
Source code repository (GitHub)	https://github.com/singhkrish2006/farmer-scheme-policy-intelligence
Project presentation (PPT)	https://github.com/singhkrish2006/farmer-scheme-policy-intelligence/blob/main/docs/Project_Demonstration.pptx
Project report (PDF)	https://github.com/singhkrish2006/farmer-scheme-policy-intelligence/blob/main/docs/Complete_Project_Report.pdf

Table 31: Project resources

The raw survey data and labelling worksheets are not published, as they contain personal information of the participants.

APPENDIX

A. Execution Steps

Step	Command	Function
1	pip install -r requirements.txt	installation of dependencies
2	python scripts/01_run_pipeline.py	cleaning and initial NLP models
3	python scripts/02_run_analysis.py	quality checks, statistics and topics
4	python scripts/07_load_database.py	database construction and 41 verification checks
5	python scripts/05_evaluate_models.py	evaluation of stored models against human labels
6	python scripts/12_compare_models.py	complete comparison of eleven models

Table 32: Execution steps

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